University of Toronto Mississauga

Mathematical and Computational Sciences
MAT102H5F - Term Test 1
Test Booklet
Duration - 110 minutes
No Aids Permitted

Surname/Family Name: ______________________

First/Given Name: ______________________

Student Number: ______________________

UTORid: ______________________

Email: ______________________

This exam contains 6 pages (including this cover page) and 5 problems. Check to see if any pages are missing and ensure that all required information at the top of this page has been filled in. You must also have the corresponding Multiple Choice Booklet, which contains the multiple choice questions needed for Question 5 (page 6). No aids are permitted on this examination. Examples of illegal aids include, but are not limited to textbooks, notes, calculators, or any electronic device.

Each part of Question 5 includes five multiple choice answers, of which only a single response is correct. All answers should be indicated on the included scantron sheet (page 6). When filling out the scantron:

- Use a #2 pencil to bubble answers (do not use ink). A #2 pencil ensures that the marks are dark enough.
- Ensure that bubbles are completely filled and make complete erasures.

Any failure to follow these instructions which results in the scanner not being able to read your work will result in a 3 mark deduction. You will not be penalized for incorrect answers.

Your solutions to Problems 1 - 4 must be written in the space provided. No additional paper may be submitted for grading. Do not, under any circumstances, write on the QR code in the top right corner of each page. **Doing so will result in you receiving a zero (0) on this test.**

Unless otherwise indicated, you are required to show your work on each problem on this exam. The following rules apply:

- **Organize your work** in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little credit.
- **Mysterious or unsupported answers will not receive full credit**. A correct answer, unsupported by calculations, explanation, or algebraic work, will receive no credit; an incorrect answer supported by substantially correct calculations and explanations might still receive partial credit.
- You may use any result mentioned explicitly in the readings, assignments, or in-class worksheets without proof. To use a result which is not covered by the aforementioned sources, you must provide a proof of the result as part of your work.

The Multiple Choice Booklet is printed one-sided, and may thus be used for rough work.

1. A set $A \subset \mathbb{R}$ is *bounded above* if there exists a $y \in \mathbb{R}$ such that for all $x \in A$, $x \leq y$.

   (i) (2 points) Give an example of a set which is bounded above.

   (ii) (6 points) For a set $A \subset \mathbb{R}$ which is bounded above, we say that $y \in \mathbb{R}$ is an *upper bound* for A if for all $x \in A$, $x \leq y$. Are there any sets $A \subset \mathbb{R}$ with a unique upper bound? Either give an example of such a set A or show that there are no sets with a unique upper bound.

   (iii) (2 points) Using the definition of bounded above as a guide, define what it means for a set $A \subset \mathbb{R}$ to be *bounded below*.

2. (i) (3 points) Define the sets $A = (0, 6) \cap \mathbb{N}$, $B = \{x \in \mathbb{Z} : x \text{ is even}\}$, and $C = \{x \in \mathbb{Z} : x^2 \leq 4\}$. Compute $A \cap B, A \cap C$ and $B \cup C$, and very clearly show that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

(ii) (7 points) Show in general that if A, B, and C are general sets, then $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

3. Suppose that P, Q and R are predicates.

   (i) (2 points) Write the truth table for $P \implies Q$.

   (ii) (8 points) Use truth tables to show that $(P \vee Q) \implies R$ is equivalent to $(P \implies R) \wedge (Q \implies R)$.

4. For each of the below statements: (1) convert the statement to an English sentence, (2) say whether the statement is true or false, and (3) prove your answer.

   (i) (5 points) $\forall m \in \mathbb{Z}, \exists n \in \mathbb{Z}, m + n$ is even.

   (ii) (5 points) $\exists m \in \mathbb{Z}, \forall n \in \mathbb{Z}, m + n$ is even.